

# Tech Assessment:

## Weather Trend Forecasting

**Candidate:** Daniel Lima de Sousa

**Role:** Dual Role

### PM Accelerator Mission

PM Accelerator is a US based company with a global reach premiering in AI learning and as a development hub, featuring award-winning AI products and mentors from top-tier companies such as Google, Meta, Apple, and Nvidia. We offer a dynamic AI PM Bootcamp, designed to empower the next generation of AI professionals through hands-on experience, mentorship, and real-world projects.

## 1. Summary and Objective

This report presents the analysis of the Global Weather Repository dataset, which contains 41 features providing daily global weather information. The objective of this project was to conduct a full Data Science lifecycle—from data cleaning to building a time series forecasting model—to predict future weather trends, with a specific focus on global temperature.

## 2. Data Cleaning & Preprocessing

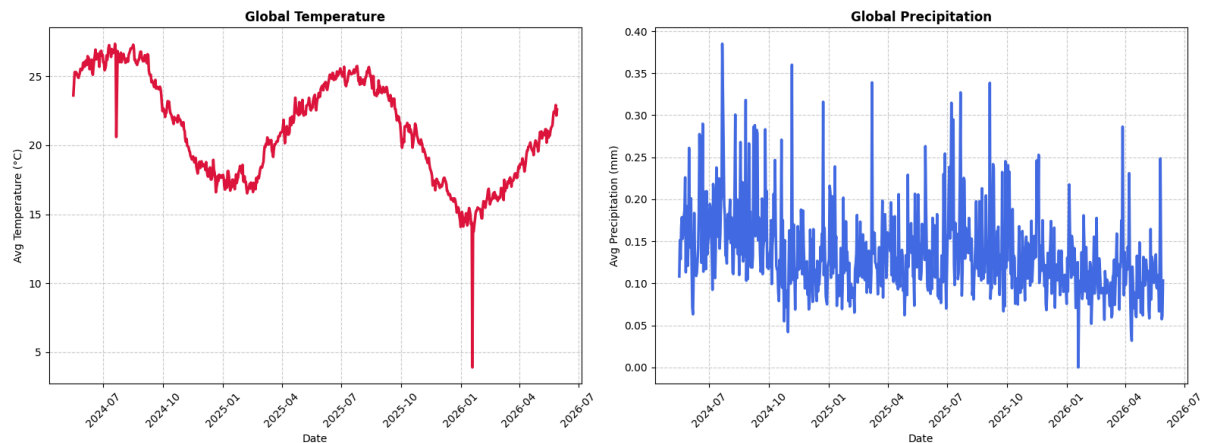
The preprocessing stage required an approach focused on the physical context of the data, avoiding "blind" statistical cuts.

- **Outlier Handling:** An initial analysis revealed that strictly using the Interquartile Range (IQR) would remove crucial data, such as days with high precipitation, which represent real extreme events rather than errors. Instead, a physics-based rule to remove catastrophic sensor failures. Impossible records such as temperatures of 79°C, wind speeds over 2900 km/h, and unrealistic atmospheric pressure (over 3000 mb) were excluded.
- **Air Quality Normalization:** For air quality variables, negative values were removed and a conservative cutoff was applied at the 99th percentile to handle long tails of anomalous pollution, resulting in the removal of approximately 5,800 rows.
- **Dimensionality Reduction:** Redundant imperial system columns (e.g., Fahrenheit, miles) were removed to prevent multicollinearity distortion.
- **Data Normalization:** To prepare the dataset for mathematical modeling, all continuous numerical variables were normalized using the *StandardScaler* from *sklearn*.

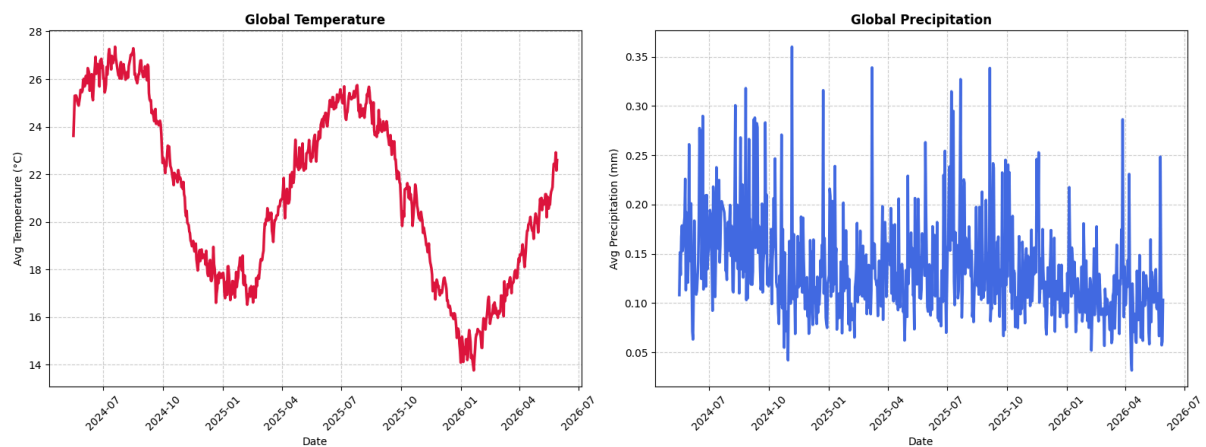
### 3. Exploratory Data Analysis (EDA)

With the sanitized data, an EDA was conducted to map global weather patterns.

**Temporal Trends** The temporal analysis confirmed the cyclical, seasonal nature of the global temperature and the high daily variance in precipitation. During temporal plotting, it identified and removed clusters of corrupted data (sudden drops of 15°C and 5°C in global averages within short 24-hour windows), ensuring the integrity of the macro trend.

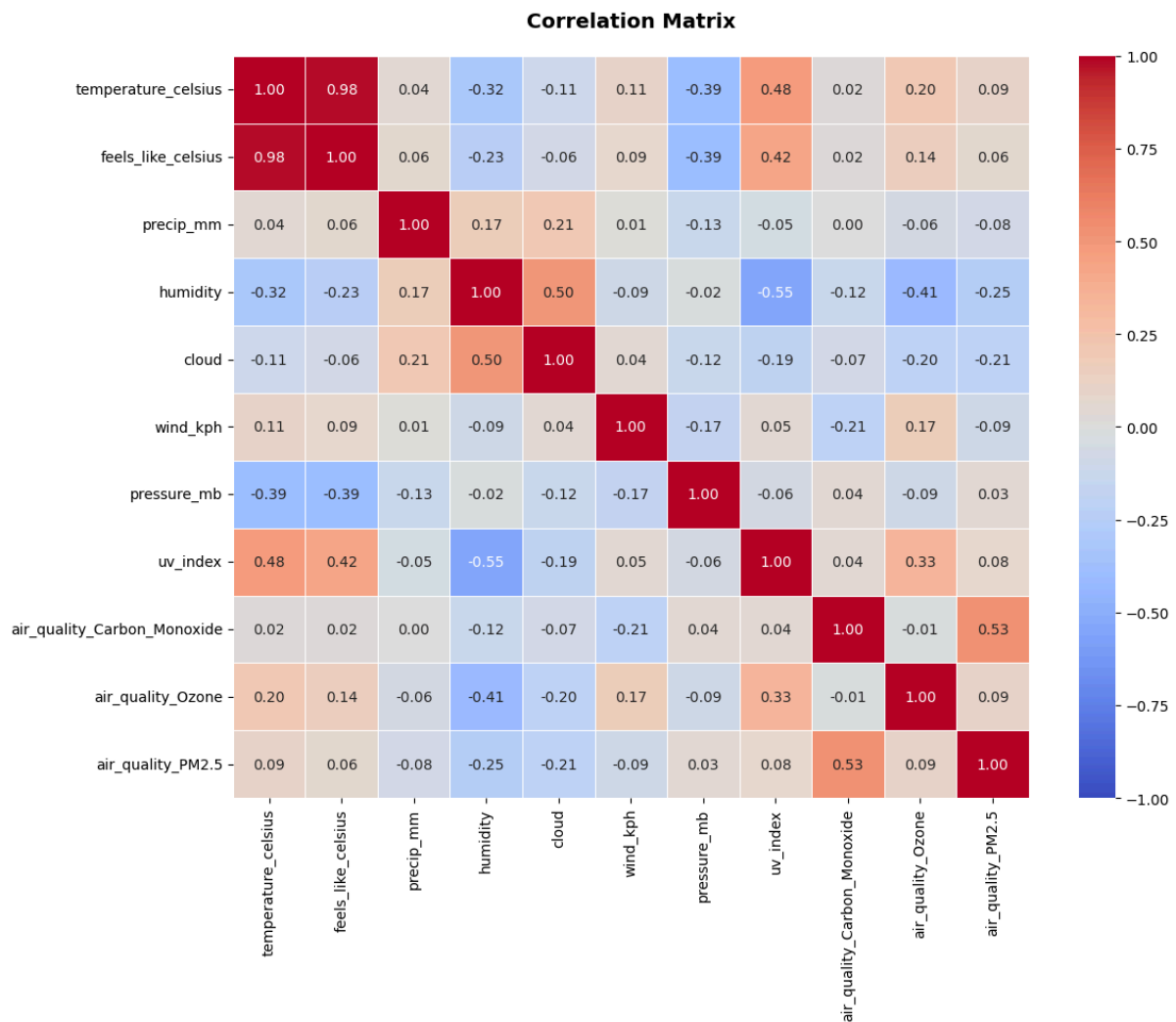


Double Line Chart with the sudden temperature drop



Double Line Chart with the outliers removed

**Correlations and Climate Interactions** To understand the underlying climate ecosystem, a Pearson correlation matrix was generated. Some logical correlations were found, such as the direct relationship between the UV Index and Temperature, and inverse relationships between Humidity, Pressure and Temperature.



**Correlation Matrix Chart**

## 4. Time Series Model Building

The core challenge was forecasting the average daily temperature. To achieve this, a Linear Regression model adapted for Time Series analysis was structured.

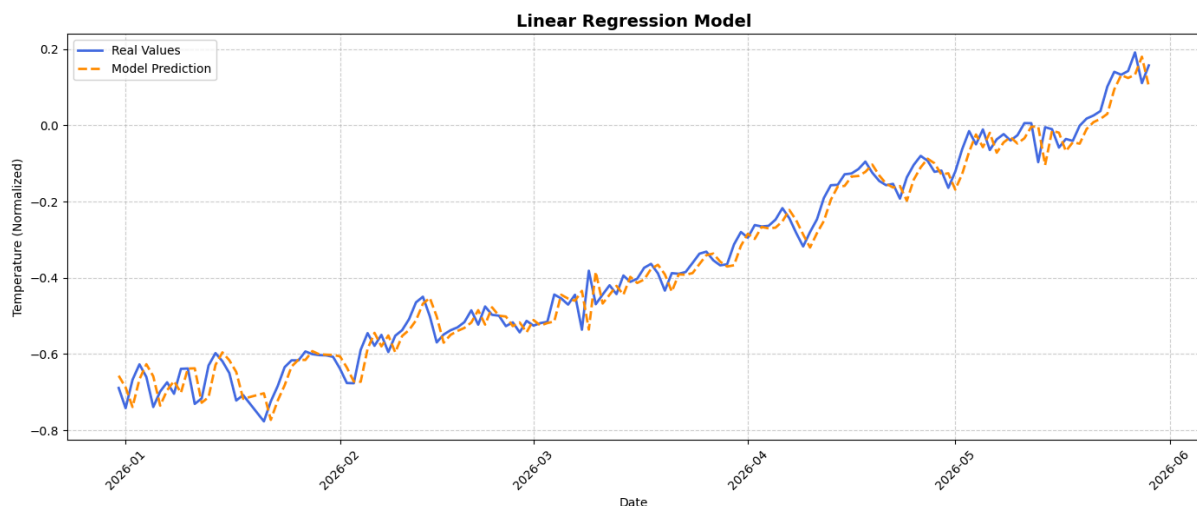
**The Role of the *last\_updated* Feature** The *last\_updated* column was converted to a *datetime* format and served as the methodological backbone of the model. The data was strictly ordered chronologically, and the train (80%) and test (20%) splits adhered to the temporal timeline to prevent future data from leaking into the past.

**Feature Engineering** The features chosen for the first model were temperature, UV, humidity, and pressure. Since there is no future data available for forecasting, the prediction was made by attempting to use the previous day's data, applying a *shift(1)* operation to all variables.

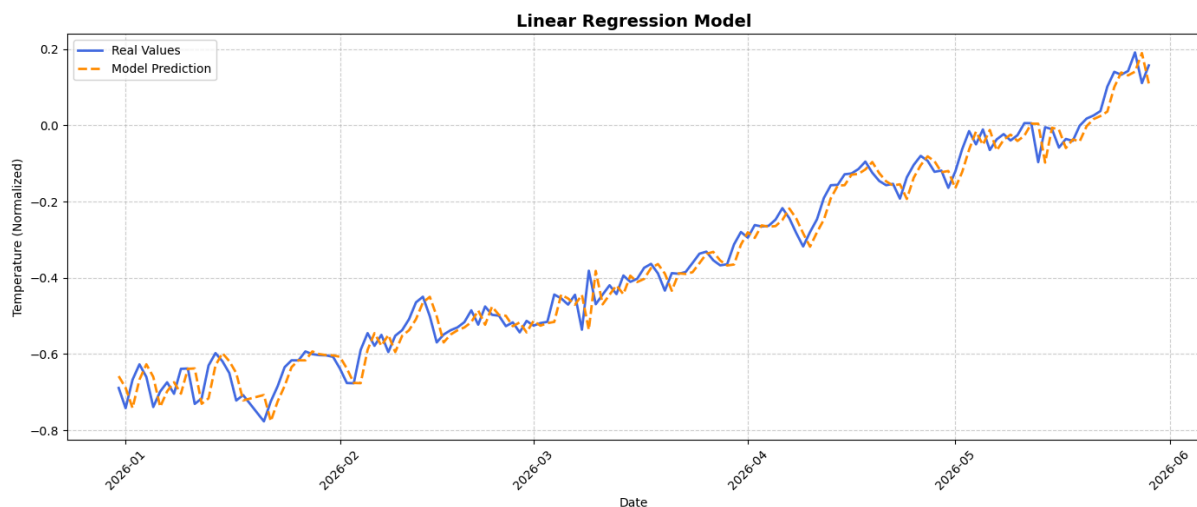
## 5. Feature Importance (Model Comparison)

To validate the predictive weight of each climate variable and improve model accuracy, three distinct configurations were created.

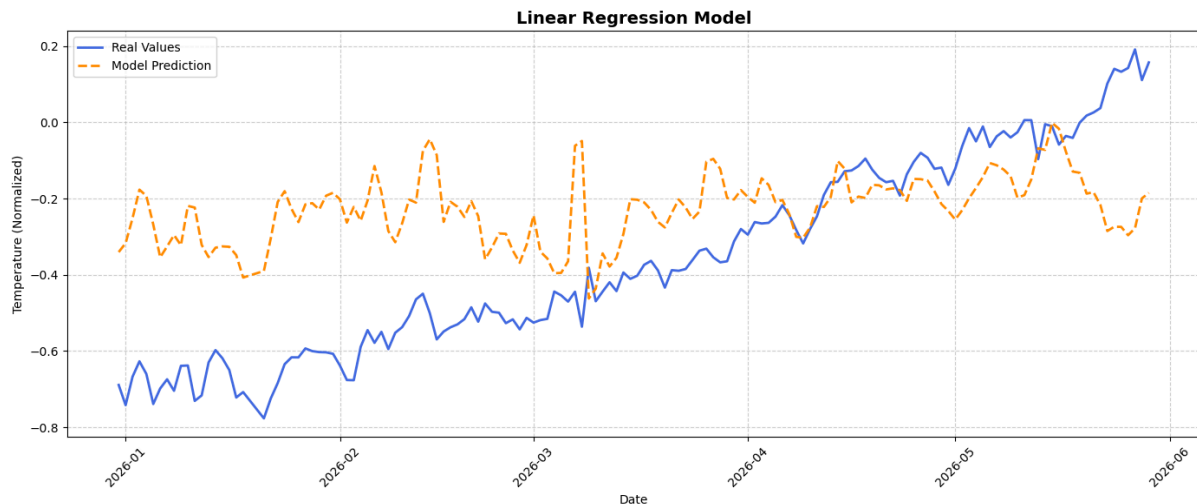
- **Model 1 (Full Multivariate):** Utilized the history (lags) of Temperature, Humidity, UV, and Pressure. It demonstrated high accuracy, closely tracking real temperature fluctuations.
- **Model 2 (Univariate):** Utilized only the Temperature history. The performance was visually identical to Model 1, demonstrating that the climate has massive inertia and past temperature is by far the dominant predictor.
- **Model 3 (Without Temperature):** The temperature history was removed from the training set, forcing the model to guess the current weather solely based on exogenous variables like pressure and humidity. The model failed to capture the trend, and the error rate spiked significantly.



**Model 1 Chart (Temperature, Humidity, UV and Pressure)**



**Model 2 Chart (Temperature Only)**



**Model 3 Chart (Humidity, UV and Pressure)**

### Performance Evaluation:

#### Model 1 (Temperature, Humidity, UV and Pressure):

- **MAE:** 0.0325
- **MSE:** 0.0017
- **RMSE:** 0.0410

#### Model 2 (Temperature Only):

- **MAE:** 0.0318
- **MSE:** 0.0016
- **RMSE:** 0.0405

#### Model 3 (Humidity, UV and Pressure):

- **MAE:** 0.2160
- **MSE:** 0.0687
- **RMSE:** 0.2621

## 6. Conclusion

Even when utilizing the exogenous variables with the highest correlation (UV Index, Humidity, and Pressure), the model struggled to learn the true underlying weather patterns. The most significant determining factor for accuracy proved to be the previous day's temperature. However, relying heavily on this variable causes the model to fall into a flawed predictive state (naive forecasting), where it simply attempts to output a value close to the previous day's temperature to predict the next day.

Potential improvements for future iterations could include: utilizing a larger, multi-year dataset to allow for more historical comparisons of the exact same dates, incorporating time-based features (such as the day of the year) and location parameters so the model can

account for regional climate patterns during its predictions, or a combination of all these approaches. The complete, reproducible code is available in the colab provided in the submission.